TEACHER GUIDE · GRADE 1 · BLOCK CODING

Build the Code

Quick facilitation notes & answer key

Goal

Students write their own short program — choosing and ordering move blocks — to make Bit reach a target number, then run it to test, and find a second program that reaches the same target. This is writing and executing code for a goal (C3.1), with a check that runs and a 'more than one way' reflection.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 and arrow/number cards so students can lay out and walk their code before writing it.

Run it (about 20 min)

1. I DO: for Build 1, count from 0 to 3 — that is 3 steps forward. Show one way to split it: 'forward 2, then forward 1.' Run it to land on 3.

2. WE DO: Exercise 1 — students fill the 2 blocks to reach 3. Exercise 2 — start at 5, reach 2 (3 steps back), predict, then run.

3. YOU DO: Exercise 3 — start at 1, reach 5 (4 steps forward); then find a second program that also reaches 5.

4. Connect: 'You wrote code to reach a goal, then ran it to test — and there can be more than one correct program.'

Answer key (modelled by the run-gate)

Ex 1 — start 0 → 3: e.g. forward 2, forward 1 (any two forward blocks summing to 3, like forward 1 + forward 2).

Ex 2 — start 5 → 2: e.g. back 2, back 1 (any two back blocks summing to 3).

Ex 3 — start 1 → 5: e.g. forward 2, forward 2; a different way: forward 1, forward 3. Both reach 5.

Why it works: the start sets Bit's number and the moves run in order; the forward/back amounts must add up to the distance from start to target.

Watch for & differentiate

Common slip: moves that overshoot or fall short — have students count the steps from start to target first, then split that into 2 blocks.

Simplify: allow one move block (e.g. 'forward 3') instead of two for students who need it.

Extend: ask for a 3-block program that reaches the target, or a program that uses both a forward and a back block.

Success indicator: the child writes a program that runs to the target and can offer a second program for Build 3.